

Jaskirat Singh | Academic CV

Final year Ph.D. Student in Computer Science | Past Intern @ Meta, Adobe Research

✉ Email: jaskirat.singh@anu.edu.au [🌐 Homepage](#) [🎓 Google Scholar](#)

Research Interests

Video Generation, Representation for Generation, Multi-modal Reasoning and Understanding.

Professional Experience

Research Scientist Intern

Creative Intelligence Lab with [Dr. Eli Shechtman](#) & [Dr. Richard Zhang](#)

Adobe Research

Dec' 25 - Present

Visiting Affiliate

VisionX group with [Prof. Saining Xie](#)

New York University

Oct' 25 - Dec' 25

Research Scientist Intern

Creative Intelligence Lab with [Dr. Eli Shechtman](#) & [Dr. Richard Zhang](#)

Adobe Research

May' 25 - Nov' 25

Research Scientist Intern

Imagine-Core (Video Generation) with [Prof. Michael Cohen](#)

Meta GenAI

Jan' 25 - April' 25

Research Scientist Intern

Imagine-Core (Video Generation) with [Prof. Michael Cohen](#)

Meta GenAI

May' 24 - Nov' 24

Research Scientist Intern

Creative Intelligence Lab with [Dr. Zhe Lin](#) & [Dr. Jianming Zhang](#)

Adobe Research

Jun' 23 - Sep' 23

Research Scientist Intern

Graphics Intelligence and Learning Lab with [Dr. Jose Echevarria](#)

Adobe Research

Jun' 21 - Dec' 21

Machine Learning Research Engineer

Mathematical Modeling for Optimizing Generative Image Content

Yahoo! Japan

Oct' 17 - Sept '18

Education

○ The Australian National University

Ph.D. in Computer Science

Supervisors: [Prof. Liang Zheng](#) and [Prof. Stephen Gould](#)

Expected

Sep' 21 - Present

○ The Australian National University

Master of Machine Learning and Computer Vision

🎓 Awarded University Medal for Exceptional Academic Excellence

GPA: 7/7

Jul' 19 - Jul' 21

○ Indian Institute of Technology, Delhi

Bachelor of Technology (B.Tech), Electrical Engineering

Specialization in Intelligent and Cognitive systems

GPA: 9.3/10

2013-2017

Selected Publications

Representation for Generation.....

1. REPA-E: Unlocking VAE for End-to-End Tuning of Latent Diffusion Transformers

Xingjian Leng*, **Jaskirat Singh***, Yunzhong Huo, Zhenchang Xing, [Saining Xie](#),
[Liang Zheng](#).

ICCV 2025

2025

2. [iREPA: What Matters for Representation Alignment: Global Information or Spatial Structure?](#)
Jaskirat Singh, Xingjian Leng, Zongze Wu, [Liang Zheng](#), [Richard Zhang](#), [Eli Shechtman](#), [Saining Xie](#). 2025
ICLR 2026
3. [Family of End-to-End Tuned VAEs for Supercharging T2I Diffusion Transformers](#)
Xingjian Leng*, **Jaskirat Singh***, Yunzhong Hou, Zhenchang Xing, Ryan Murdock, Ethan Smith, Rebecca Li, [Saining Xie](#), [Liang Zheng](#). 2025
🏆 >5x faster and Efficient T2I training over Flux VAE

RL-Driven Reasoning Agents.....

4. [R2E-Gym: Procedural Environment Generation and Hybrid Verifiers for Scaling Open-Weights SWE Agents](#)
Naman Jain*, **Jaskirat Singh***, Manish Shetty, [Liang Zheng](#), [Koushik Sen](#), [Ion Stoica](#). 2025
🏆 Top 5 papers COLM 2025, NeurIPS Workshop 2025
5. [DeepSWE: Training a Fully Open-sourced State-of-the-Art Coding Agent by Scaling RL](#)
Michael Luo*, **Jaskirat Singh***, Naman Jain*, Sijun Tan*, Colin Cai*, et al. 2025
🏆 State-of-the-art Open-weight Coding Agents using pure RL
6. [OpenDevin: An Open Platform for AI Software Developers as Generalist Agents](#)
Best LLM Agent for 🏆 SWE-Bench Verified & 🏆 SWE-Bench-Lite. 2024
🏆 ICLR 2025 Oral

Controllable Image and Video Generation.....

7. [Negative Token Merging: Image-based Adversarial Feature Guidance](#)
Jaskirat Singh, [Lindsey Li](#), [Weijia Shi](#), [Ranjay Krishna](#), [Yejin Choi](#), [Pang Wei Koh](#), [Michael F. Cohen](#), [Stephen Gould](#), [Liang Zheng](#), [Luke Zettlemoyer](#). 2024
🏆 Image based classifier-free guidance to improve diversity, quality in few lines of code
8. [Storybooth: Training-free Multi-Subject Consistency for Improved Visual Storytelling](#)
Jaskirat Singh, Junshen Kevin Chen, Jonas Kohler, [Michael Cohen](#). 2025
ICLR 2025
9. [SmartMask: Context Aware High-Fidelity Mask Generation for Fine-grained Object Insertion and Layout Control](#)
Jaskirat Singh, [Jianming Zhang](#), [Qing Liu](#), [Cameron Smith](#), [Zhe Lin](#), [Liang Zheng](#). 2024
CVPR 2024 , US Research Patent (Adobe)
10. [IMPUS: Image Morphing with Perceptually-Uniform Sampling Using Diffusion Models](#)
Zhaoulun Yang, Zhengyang Yu, **Jaskirat Singh**, Zhiwei Xu, Jing Zhang, Dylan Campbell, Peter Tu, and [Richard Hartley](#). 2023
ICLR 2024
11. [Divide, Evaluate, and Conquer: A Compositional Perspective on Evaluating and Improving Text-to-Image Alignment](#)
Jaskirat Singh, and [Liang Zheng](#). 2023
NeurIPS 2023
12. [High-Fidelity Guided Image Synthesis with Latent Diffusion Models](#)
Jaskirat Singh, [Stephen Gould](#), and [Liang Zheng](#). 2023
CVPR 2023
13. [Paint2Pix: Interactive Painting based Progressive Image Synthesis and Editing](#)
Jaskirat Singh, [Liang Zheng](#), [Cameron Smith](#), and [Jose Echevarria](#). 2022
ECCV 2022

14. [Intelli-Paint: Towards Developing Human-like Painting Agents](#)
Jaskirat Singh, Cameron Smith, Jose Echevarria, and Liang Zheng. 2022
ECCV 2022, US Research Patent (Adobe)
15. [Combining Semantic Guidance and Deep Reinforcement Learning For Generating Human-Level Paintings](#)
Jaskirat Singh, and Liang Zheng. 2021
CVPR 2021

Honors and Achievements

-  **Awarded University Medal** for academic excellence at the Australian National University.
-  **Awarded Chancellors Letter of Commendation** at the Australian National University.
- Awarded **ANU Computer Science Summer Research Grant** (\$10k).
- Our project "Connected Stories of Australia" has been awarded as the **best innovative design project** by the **National Museum of Australia**.
- **Won national hackday at Yahoo Japan**, among 54 competing teams from all across Japan, for developing a real-time application for **facial attribute modification using reversible GANs**.
- Received **IIT Delhi Merit Award & Scholarship** for outstanding academic performance.
- Secured **All India Rank 128 in IIT-JEE** among 1.4 million aspirants appearing for the exam.
-  **Silver Medalist at National FIDE Rated Chess Tournament.**

Referees

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| ◦ Prof. Saining Xie - Assistant Professor |  NYU, AMI Labs |
| ◦ Prof. Michael Cohen - Director of Computational Photography |  Meta GenAI, UW |
| ◦ Dr. Eli Shechtman - Senior Principal Research Scientist |  Adobe Research |
| ◦ Prof. Liang Zheng - Associate Professor |  Australian National University |
| ◦ Prof. Stephen Gould - Senior Professor |  ANU, Amazon Research |
| ◦ Prof. Marcus Hutter - Senior Research Scientist |  Google Deepmind |
| ◦ Dr. Jianming Zhang - Senior Principal Research Scientist |  Adobe Research |
| ◦ Dr. Zhe Lin - Senior Principal Research Scientist |  Adobe Research |